Northern University of Business and Technology Khulna
Department of Computer Science and Engineering

[Project Work Plan]

Project Title:
Development of a Car Racing Game

Submitted by:
Sajedur Jaman
11240321742
sajedurjaman@gmail.com

Course Name: Software Development I
Course Code: CSE 2216
Section: 4B

Submitted to:
Vashkar Kar
Lecturer,
Department of Computer Science and Engineering,
Northern University of Business and Technology Khulna

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Project Title: Car Simulator

Course: Software Development Lab

Programming Language: Java (Swing & AWT)

10-Week Work Plan for Car Simulator Project

Week	Phase
1	Project Planning & Concept Finalization
2	Requirements Analysis & System Design
3	Environment Setup & Project Initialization
4	Game Window & Core Loop Setup
5	Road System & Player Car
6	AI Traffic & Collision Detection
7	Environment & Parallax Scenery
8	Scoring, HUD & Win/Game Over Screens
9	Error Handling & Testing
10	Final Testing, Documentation & Submission

Week 1: Project Planning & Concept Finalization

- Select project topic and objectives.
- Define core gameplay mechanics, win condition, and project scope.

Prepare initial project plan

Week 2: Requirements Analysis & System Design

- Gather functional requirements for the road, traffic, and scoring systems.
- Design the object-oriented class architecture (Car, Road, Environment, GasItem).
- Create basic wireframes for the game window and HUD overlay.

Week 3: Environment Setup & Project Initialization

- Install JDK and configure the IDE.
- Configure the Java Swing/AWT environment.
- Create the project structure and the CarRacingGame JFrame entry point.

Week 4: Game Window & Core Loop Setup

- Implement the GamePanel class with the core update/repaint cycle.
- Set up the javax.swing.Timer to drive a ~60 FPS game loop.
- Establish the 900x700 game window and base rendering pipeline.

Week 5: Road System & Player Car

- Implement the 3-lane scrolling Road class with curve physics.
- Implement the abstract Car class and PlayerCar with lane-switching and speed boosting.
- Wire up Left/Right Arrow and Space key controls.

Week 6: AI Traffic & Collision Detection

- Implement EnemyCar with randomized colors, speeds, and directions.
- Add AABB collision detection between the player and enemy cars.

- Implement GasItem pickups with spin/glow animation and speed boost effect.

Week 7: Environment & Parallax Scenery

- Implement the Environment manager with Tree, Bush, Mountain, Lake, and Park classes.
- Add parallax scrolling so background layers move at different speeds for depth.
- Implement animated Bird and Cloud elements and the dynamic dawn-to-dusk sky.

Week 8: Scoring, HUD & Win/Game Over Screens

- Implement the scoring system (+10 per car passed, +5 per gas canister collected).
- Build the HUD overlay showing live score, speed, and distance progress.
- Implement the Game Over and You Won screens with restart via the R key.

Week 9: Error Handling & Testing

- Test collision detection, lane switching, and road curve physics.
- Handle edge cases (rapid lane switches, simultaneous boosts, win/loss timing).
- Fix bugs and optimize rendering performance for smooth 60 FPS gameplay.

Week 10: Final Testing, Documentation & Submission

- Perform complete system testing across all gameplay scenarios.
- Prepare project documentation and screenshots.
- Finalize and submit the project.